

EXPERIENCE

Research Intern, Microfluidics Lab - Jun 2024 - Aug 2024

- Wrote a grading script that runs student submissions in isolated containers and records time and memory used for each test
- Rebuilt the volunteer scheduling tool so coordinators can swap shifts from their phones and see open slots

Teaching Assistant, Statistics Department - Aug 2024 - May 2025

- Trained a small convolutional network to sort recycling photos taken at the two dining hall bins every week

Machine Learning Reading Group

- Designed a printed circuit board for a soil moisture logger and wrote firmware that wakes up once an hour
- Campus Bus Tracker - React and Flask web app used by about 200 students.

- Interviewed twenty transit riders about late buses and summarized their answers in a two-page city memo

Presented the results at the Undergraduate Research Symposium in April.

Honors & Awards

- Dean's List for five semesters

ACTIVITIES

Software Engineering Intern, Prairie Analytics

- Analyzed survey responses from 1,200 undergraduates about rent and commute costs with Python, SQL
- Designed, built and tested a line-following robot for the intro course
- Built an inventory dashboard that the campus food pantry checks before every delivery in Champaign, IL
- Translated onboarding documents from Spanish to English for a legal clinic
- Graded weekly programming assignments for 180 students and wrote the autograder test suites in C++
- Led two review sessions before each midterm exam
- Ran weekly code reviews for the twelve members of the robotics team and kept notes on recurring bugs;
- Mentored two new members through their first pull requests
- Rewrote the student radio station's scheduling app so the hosts can trade time slots using React, Node.js
- Python, SQL, Tableau

Data Analyst, City of Champaign

Analyzed survey responses from 1,200 undergraduates about rent and commuting costs with Python, SQL

Designed, built and tested a line-following robot for the intro course

Built an inventory dashboard that the campus food pantry checks before every shipment in Champaign, IL

Research Intern, Microfluidics Lab

Rewrote the student radio station's scheduling app so that hosts can trade their slots using React, Node.js

Python, SQL, Tableau

实习经历

数据分析实习生，图像模型项目 2025.06 - 2025.08

- 为同学整理课程资料，训练五个模型，为课程设计测试，参与项目设计，维护项目网站，并标注四千张图像
- 分析模型测试数据，维护实验系统，为同学整理课程资料，参与项目设计，整理标注规范，并训练图像模型

研究助理，图像标注实验室 2024.09 - 2025.05

- 负责实验数据的整理与分析，用 Python 训练图像模型，整理标注规范，维护项目网站，并为课程设计测试

校园活动

组织同学参与编程课程，每周撰写实验报告，维护实验系统，用 Python 训练图像模型，并按照规定标注图像

负责实验数据的整理与分析，用 Python 训练图像模型，按照规定标注图像，维护项目网站，并撰写项目报告

组织同学参与编程课程，用 Python 训练图像模型，标注四千张图像，参与项目设计，并为同学整理课程资料

- 撰写项目报告，设计与测试模型，组织同学参与编程课程，整理标注规范，训练五个模型，并维护项目网站

荣誉

- 校级优秀学生奖学金

EXPERIENCE

Software Engineering Intern, Prairie Analytics

- Rewrote the student club website in Next.js so officers can post their events without asking the webmaster

Caterpillar

Data Science Intern

- Trained a gradient boosting model to predict which books students will request next term at the main library

Accomplishments

- Reviewed pull requests for a student-run open-source project and wrote contributor guidelines of the club

COMPETITIONS

- Organized a weekend hackathon for 150 students with prize sponsors from four local companies in Urbana
PantryPal

- Tools: Python, SQL, Airflow, dbt, Docker, Kubernetes, Terraform, AWS Lambda, Redshift, Looker, Git, Jira

Teaching Assistant, Statistics Department

- Maintained the equipment checkout system that the photography club and two other student groups rely on
Research Intern, Microfluidics Lab

- Set up the tutoring schedule and the waitlist form that the students in the intro physics course signed up for
Volunteer Coordinator, Eastern Illinois Foodbank - Jun 2024 - Aug 2024

PROJECTS

- Wrote unit tests for the parser.
- Added a nightly job that checks the backups.
- Mapped bike lane gaps around campus with QGIS and presented the map to the facilities office last spring
- A web app that shows live bus positions to about 200 students.
- Designed the logger board and wrote the firmware that wakes up once an hour to save power on the farm
- Custom firmware that logs soil moisture every hour on battery power.
- Interviewed twenty transit riders about late buses and summarized their answers in a two-page city memo
- PantryPal is an inventory tracker for the campus food pantry.

ACTIVITIES

Tools: SolidWorks, MATLAB, LabVIEW, COMSOL, Arduino, ImageJ, Excel, Python, Simulink, KiCad, Altium
Research Intern, Microfluidics Lab

Built a self-service sign-in portal for the campus makerspace that all 300 student members now use to log in

Mentored two new members through their first pull requests

Ran gel electrophoresis for a plant genetics lab in Urbana and kept the shared sample database consistent

12 students mentored through their first research projects

Technologies: React, TypeScript, Node.js, Express, PostgreSQL, Redis, Docker, GitHub Actions, Jest, Bash

Finalist, Illinois Innovation Prize

EXPERIENCE

• Tools: SolidWorks, MATLAB, LabVIEW, COMSOL, Arduino, ImageJ, Excel, Python, Simulink, KiCad, Altium
Beckman Institute, University of Illinois

Undergraduate Researcher - Sep 2025 - Present

Skills: Python, SQL, Airflow, dbt, Docker, Kubernetes, Terraform, AWS Lambda, Redshift, Looker, Git, Excel
Caterpillar Inc., Peoria

Relevant Coursework: Operating Systems, Linear Algebra, Computer Vision, Database Systems, Algorithms
Dean's List, James Scholar

Profiled the image preprocessing step of a crop disease classifier and cut its runtime in half on the lab GPU
40% faster nightly builds after moving the test suite to parallel runners

- Organized a hackathon for 150 students with sponsors from four local companies and ran the judging day
 - 1,200 survey responses coded for a campus housing study
 - Built the badge scanner that the volunteers at the food pantry on Green Street use whenever they check in
- GitHub Campus Expert Program